

27 Which sequence of events could lead to mass flow in a phloem sieve tube?

- 1 Sucrose moves into a sieve tube element.
- 2 Protons (H^+) and sucrose molecules move into a companion cell through a co-transporter protein.
- 3 A very high hydrostatic pressure is produced.
- 4 The water potential of the sieve tube element decreases.
- 5 Protons (H^+) are actively pumped out of a companion cell.
- 6 Water moves down a water potential gradient by osmosis.

A 1 → 2 → 5 → 6 → 4 → 3

B 3 → 1 → 2 → 5 → 6 → 4

C 3 → 5 → 1 → 2 → 4 → 6

D 5 → 2 → 1 → 4 → 6 → 3